WEEK 3 · 23–26 NOVEMBER 2026

# AI-assisted classification and mapping

Training, inference and honest maps — the practical half of GeoAI

Week 3 · Session 2 · practical: 20_geoai_deep_learning.ipynb

LEARNING OBJECTIVES

# What you will be able to do

- Prepare a labelled dataset from WorldCover pure pixels, and know why purity matters
- Build a **spatially blocked** split with a buffer that matches your patch size
- Read a training curve: early stopping, overfitting, underfitting, unstable validation
- Run **tiled inference** over a whole scene and export a Cloud-Optimised GeoTIFF
- Produce and interpret a confidence map and an MC-dropout uncertainty map
- Diagnose the usual failures: NaNs in patches, class collapse, memory, slow CPUs

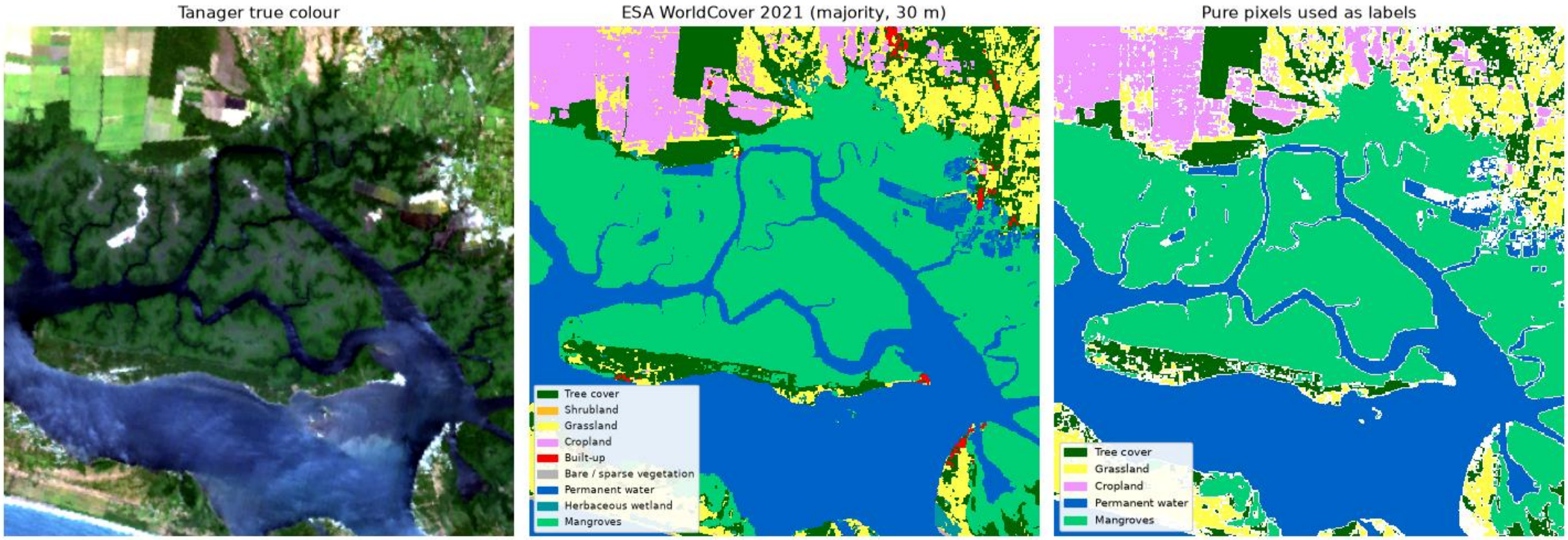
# 1. The dataset

Labels, purity and class balance

# What we train on, and how the labels are cleaned

- One **400 × 400 pixel window** (12 × 12 km) of the El Salvador mangrove coast
- **352 good bands** — water-vapour windows, SWIR edge and sub-400 nm bands already removed
- Labels: **ESA WorldCover 2021** at 10 m, aggregated to the Tanager 30 m grid
- Five classes with enough pure pixels: tree cover, grassland, cropland, water, mangroves
- **purity = 9**: all nine 10 m cells inside the 30 m pixel agree — mixed pixels are dropped
  - Cost: labels become sparse along boundaries — mangrove edges, field margins, creeks

# The window, the labels and what survives the purity filter




*True colour · WorldCover classes · pure-pixel labels used for training*

# 2. Splitting the data

The step that decides whether your accuracy is real

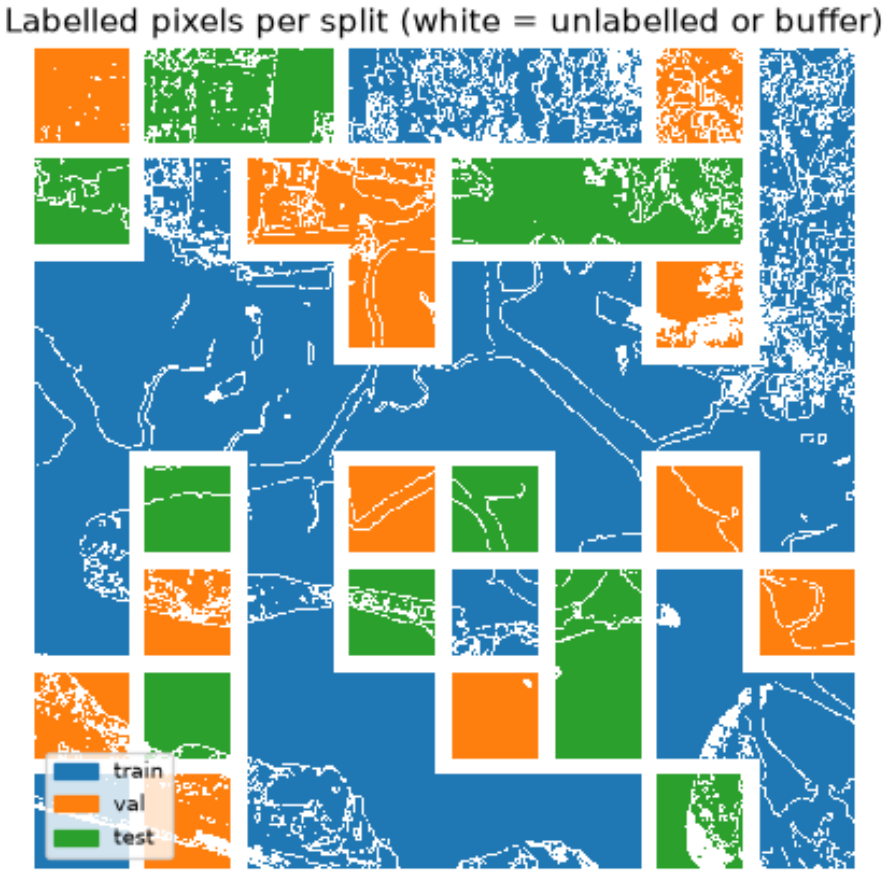
THE MOST COMMON MISTAKE IN THE LITERATURE

# A random pixel split does not measure generalisation. It measures memorisation.

Neighbouring hyperspectral pixels are nearly identical. With 9 × 9 patches, a test patch can literally contain pixels the network trained on.

# Blocks, buffers and what gets thrown away

- Cut the window into **50 × 50 px blocks** (1.5 × 1.5 km)
- Assign blocks 60 / 20 / 20 to train / validation / test
- Patch radius R = 4, so drop any pixel whose 9 × 9 window crosses another split
- **The buffer must grow with the patch size** — change R, change the buffer
- Validation drives early stopping; the test set is touched **once**




*Block split with a 4-pixel buffer between splits*

# 3. Training

Two models, one baseline, and how to read the curves

# What we run, and why in this order

- **Random Forest** on the full 352-band spectrum — the baseline, trains in ~2.5 s
- **1D-CNN** on spectra — 42 309 parameters, ~41 s on CPU, stopped after 12 epochs
- **HybridSN-style 3D→2D CNN** on 9 × 9 × 30 PCA patches — 265 349 parameters, ~35 s
- PCA is fitted on **training pixels only**; fitting on everything leaks test information
- 30 components keep **99.99 %** of the training variance, so the patch model loses almost nothing
  - Always report the baseline next to the deep model — it is the honest comparison

FOUR LINES THAT MATTER

# The ingredients that make it train on small data

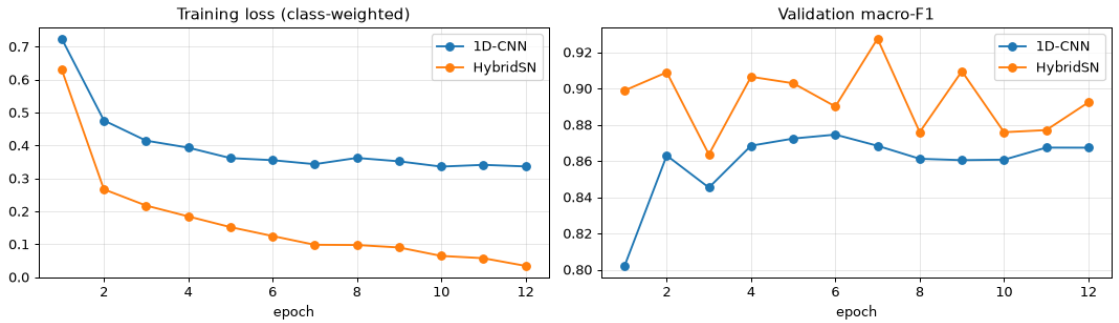
```
# class-weighted loss: rare classes count as much as common ones
w = N / (K * counts)                          # w_k = N / (K · N_k)
loss = nn.CrossEntropyLoss(weight=torch.tensor(w, dtype=torch.float32))

# standardise with TRAIN statistics only — never the test set
band_mean, band_std = Xtr.mean(0), Xtr.std(0) + 1e-6

# early stopping: keep the weights with the best validation macro-F1
if val_f1 > best_f1:
    best_f1, best_state = val_f1, copy.deepcopy(model.state_dict())
```

Class weighting, standardisation, early stopping — the rest is plumbing

# Reading the training curves




*Training loss and validation macro-F1 per epoch*

- Training loss falls, validation macro-F1 rises, then flattens — stop there
- **Overfitting:** validation F1 turns down while training loss keeps falling
- **Underfitting:** both stay flat and poor — model too small or learning rate wrong
- **Noisy validation curve:** the validation set is small or unrepresentative
- Both models stopped after 12 epochs here

# 4. Results, maps and shipping

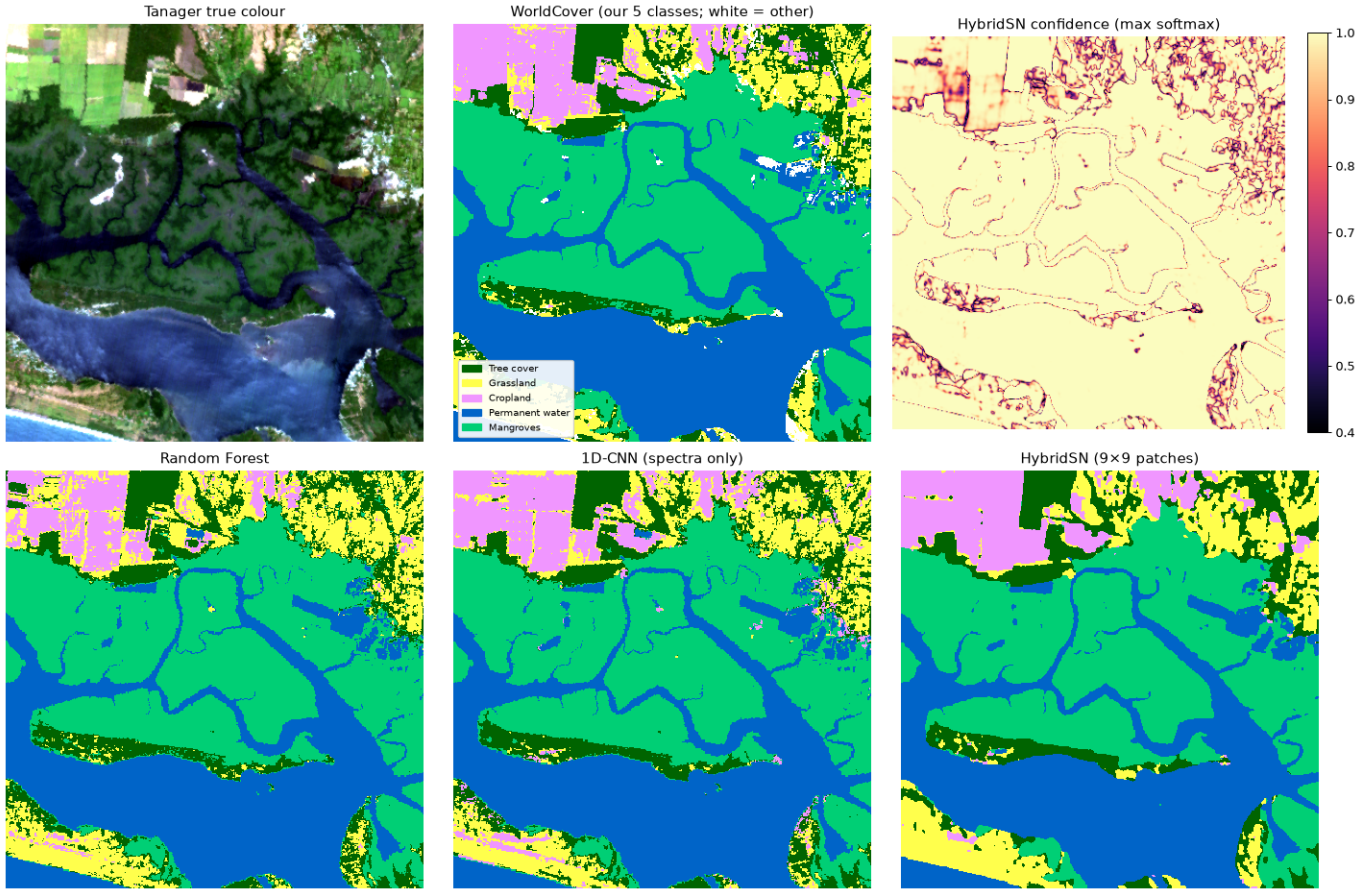
Comparing models, then scaling to a whole scene

MEASURED, NOT QUOTED

# Per-class F1 on the spatially independent test blocks

| | Random Forest | 1D-CNN | HybridSN |
|---|---|---|---|
| Tree cover | 0.836 | 0.830 | 0.882 |
| Grassland | 0.646 | 0.720 | 0.804 |
| Cropland | 0.851 | 0.940 | 0.960 |
| Permanent water | 0.995 | 0.994 | 0.994 |
| Mangroves | 0.979 | 0.974 | 0.983 |
| macro F1 | 0.861 | 0.892 | 0.925 |
| overall accuracy | 0.925 | 0.940 | 0.959 |

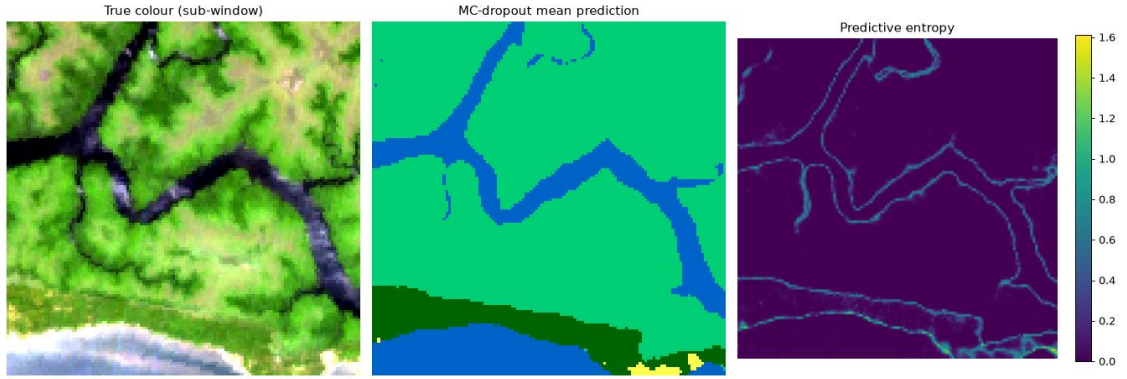
# From model to map




*True colour · WorldCover · HybridSN confidence · the three model maps*

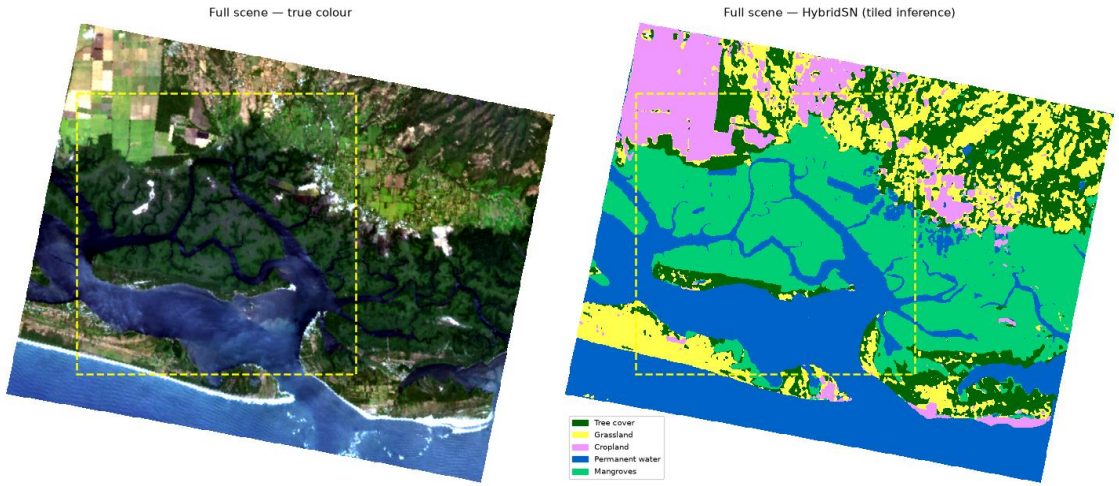
# MC-dropout: a second opinion on uncertainty

- Keep dropout **on** at prediction time and run the model several times
- Different neurons drop each pass, so predictions differ
- Summarise with predictive entropy $H = -\Sigma\ \bar{p} \log \bar{p}$ (0 = certain)
- 10 passes over 16 900 pixels took 32 s here
- High entropy marks boundaries and out-of-legend materials




*Mean prediction, confidence and MC-dropout entropy*

# Tiled inference over the full scene




*HybridSN applied to the whole Tanager scene, tile by tile*

- The full cube here is ~650 × 785 px × 352 bands ≈ **0.75 GB** in memory
- Read **one strip of rows** at a time, with a halo of R rows above and below
- Predict the core rows, write them out, move on — memory stays bounded
- Full scene in **88 s** on CPU, exported as a COG plus a confidence layer
- Yellow box = the training window; everything else is extrapolation

DOMAIN SHIFT, MEASURED

# Does the model travel?

| Test | Overall accuracy | Macro F1 |
|---|---|---|
| Test blocks (same window) | 0.959 | 0.925 |
| Same scene, outside the window | 0.843 | 0.792 |
| Ho Chi Minh City — HybridSN | 0.335 | — |
| Ho Chi Minh City — 1D-CNN | 0.334 | — |
| Ho Chi Minh City — Random Forest | 0.261 | — |

# Troubleshooting: what actually goes wrong

- **NaNs in a patch** — no-data or cloud inside the window: mask, or fill with the PCA mean (zero)
- **Class collapse** (everything one class) — imbalance: check the class weights and the loss
- **Validation F1 stuck at chance** — labels and features misaligned, or learning rate too high
- **MemoryError on patches** — build patches with a sliding-window *view*, predict in batches
- **Training is slow** — `torch.set_num_threads(4)`, fewer PCs, smaller patches, fewer pixels
- **Test accuracy suspiciously high** — check the split: leakage before celebration

PRACTICAL

# Over to you

▶ `20_geoai_deep_learning.ipynb` · `120 min`

- Sections 2–3: build the dataset and the blocked split — check the buffer
- Sections 4–6: train the RF baseline, the 1D-CNN and the HybridSN patch model
- Sections 7–9: compare honestly, map the window, add confidence and MC-dropout entropy
- Sections 10–11: tiled inference, export the COG, then the domain-shift test
- Exercises: patch size R = 2 vs 6 · random split vs block split · learning curve with 1–30 % of labels

KEY TAKEAWAYS

# Recap

- Labels are a **map, not truth** — filter for purity and say what that costs
- **Spatial blocks plus a buffer** matched to the patch size; the test set is touched once
- Report the **baseline**: 0.861 macro F1 (RF) against 0.925 for the patch CNN
- Maps need **confidence and uncertainty layers**, and someone to look at them
- **Tiled inference** scales the same code from one window to an archive
- Accuracy is **local**: 0.959 in the test blocks, 0.84 outside the window, ~0.3 in another country

**Next: Session 3 — Validation methods and accuracy assessment (notebook 21)**

# References and further reading

Roy, S.K. et al. (2020). HybridSN: exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification. IEEE GRSL 17(2), 277–281.

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Breiman, L. (2001). Random forests. Machine Learning 45, 5–32.

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